Exercise Class 1 — Consumer Theory (MWG Ch. 1–3)

Advanced Microeconomics 1 · Thursday 8 October, 11:00–13:00

Format (120 min). 60 min: assigned students present Problems 1–5 at the board (assignments posted Monday 5 October). 30 min: instructor works the hard item (Problem 6). 30 min: timed prelim-style question (Problem 7), closed book, ungraded, self-marked against the sketch afterwards.

Part I — Board problems

Problem 1 Rational preferences and their limits

(a) Let $\succsim$ be rational. Prove that $\succ$ is transitive and irreflexive, and that $\sim$ is an equivalence relation.

(b) Give an example of a complete but intransitive relation on a three-element set, and exhibit an explicit money pump against an agent with those strict preferences.

(c) Lexicographic preferences on $\mathbb{R}^2_+$: $x \succsim y$ iff $[x_1 > y_1]$ or $[x_1 = y_1$ and $x_2 \geq y_2]$. Show they are rational and strongly monotone; show they are not continuous; sketch why no utility representation exists (map each x_1 to a rational number in a suitably chosen interval and count).

(d) For finite X and rational $\succsim$, verify that $u(x) = \#\{y \in X : x \succsim y\}$ represents $\succsim$.

Problem 2 Choice structures and WARP on a finite domain

Let $X = \{a, b, c\}$ and $\mathcal{B} = \{\{a, b\}, \{b, c\}, \{a, c\}\}$ (pairs only), with

$$C(\{a, b\}) = \{a\}, \qquad C(\{b, c\}) = \{b\}, \qquad C(\{a, c\}) = \{c\}.$$

(a) Verify that this choice structure satisfies WARP.

(b) Show that no rational preference rationalizes it.

(c) Now add $\{a, b, c\}$ to $\mathcal{B}$. Show that *no* specification of $C(\{a, b, c\})$ preserves WARP. Reconcile with MWG Proposition 1.D.2.

Problem 3 The CES workout

Let $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$, $\rho < 1$, $\rho \neq 0$; write $\sigma = 1/(1 - \rho)$ and $P = (p_1^{1-\sigma} + p_2^{1-\sigma})^{1/(1-\sigma)}$.

(a) Derive $x(p, w)$ and $v(p, w)$ from the UMP.

(b) Derive $h(p,u)$ and $e(p,u)$ from the EMP.

(c) Verify all four duality identities: $e(p,v(p,w)) = w$; $v(p,e(p,u)) = u$; $h(p,u) = x(p,e(p,u))$; $x(p,w) = h(p,v(p,w))$.

(d) Verify Roy's identity and Shephard's lemma by direct differentiation.

(e) Recover Cobb–Douglas as $\sigma \to 1$ and Leontief as $\sigma \to 0$ (limits of $e(p,u)$ suffice).

(f) *The two accounting laws at work.* For a general demand system satisfying homogeneity of degree zero and Walras' law, differentiate the two identities to derive **Engel aggregation** $\sum_\ell p_\ell\, \partial x_\ell/\partial w = 1$ and **Cournot aggregation** $\sum_\ell p_\ell\, \partial x_\ell/\partial p_k = -x_k$. Conclude that not every good can be inferior. Then verify both identities explicitly for the CES system of part (a).

Problem 4 Quasilinear preferences: the welfare-exact case

Let $u(x_0, x_1) = x_0 + \phi(x_1)$ with $\phi' > 0 > \phi''$, $p_0 = 1$, and w large enough for interior solutions.

(a) Derive $x_1(p_1)$ (note: independent of w), $v(p_1, w)$, $e(p_1, u)$, $h_1(p_1)$.

(b) Show $h_1(p_1, u) = x_1(p_1)$ for every u: Hicksian and Walrasian demand coincide.

(c) For a price change $p_1^0 \to p_1^1$, prove CV = EV = ΔCS = $\int_{p_1^1}^{p_1^0} x_1(p)\, dp$.

(d) Exactly which step fails when wealth effects are restored? Point to the term in the Slutsky equation.

Problem 5 Slutsky by hand, once in your life

For the CES demand system of Problem 3 with $\sigma = 2$, $p = (1,1)$, $w = 2$:

(a) Compute the 2×2 Slutsky matrix $S(p,w) = D_p x + D_w x\, x^\top$ numerically.

(b) Verify symmetry, negative semidefiniteness (compute eigenvalues or check $s_{11} \le 0$ and $\det S = 0$), and $Sp = 0$.

(c) Confirm $S = D_p h$ evaluated at $u = v(p,w)$.

(d) With $L = 2$, show from NSD + singularity alone that $s_{12} \ge 0$: two goods must be net substitutes.

Part II — Instructor-worked hard item

Problem 6 Afriat by hand

Consider $T = 3$ observations on two goods:

t	p_1^t	p_2^t	x_1^t	x_2^t
1	2	1	3	2
2	1	2	2	3
3	1	1	4	1

(a) Build the 3×3 expenditure cross-matrix $E_{ts} = p^t \cdot x^s$ and the direct revealed preference relations R^D (chosen bundle weakly cheaper comparison) and P^D.

(b) Compute the transitive closure and check GARP.

(c) If GARP holds, find Afriat numbers (U_t, λ_t) solving $U_s \le U_t + \lambda_t \, p^t(x^s - x^t)$ for all s, t, and write down the rationalizing utility $u(x) = \min_t\{U_t + \lambda_t \, p^t(x - x^t)\}$. If it fails, exhibit the violating cycle(s) and compute the CCEI by hand (find the budget scaling at which the last cycle just breaks).

(d) Now replace the third observation by $x^3 = (2, 2)$ (same prices). Re-check GARP; construct (U_t, λ_t) — try $\lambda_t = 1$ — and the rationalizing $u(x) = \min_t\{U_t + \lambda_t \, p^t(x - x^t)\}$; evaluate u at the three bundles and confirm each is optimal in its own budget set.

Part III — Timed prelim question (30 minutes, closed book)

Problem 7 Reconstructing the square from v

A consumer's indirect utility is

$$v(p_1, p_2, w) = \frac{w}{p_1^{\alpha} \, p_2^{1-\alpha}}, \qquad \alpha \in (0, 1).$$

(a) Verify v has the properties required of an indirect utility function (h.d.0, monotonicity, quasi-convexity may be asserted with a one-line reason).

(b) Recover Walrasian demand via Roy's identity.

(c) Recover $e(p, u)$ by inverting, and Hicksian demand via Shephard's lemma.

(d) Compute the Slutsky matrix and verify symmetry, NSD, and $Sp = 0$.

(e) Which familiar preferences generated this v? Justify.